

Binod Bishwakarma

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github.com/binodbishwakarama-max | Portfolio

SUMMARY

Full-Stack and AI/ML Engineer with 4+ deployed systems across AI, full-stack, and ML. Specializes in embedding LLM systems into real-world workflows with a focus on latency, reliability, and deployment under constraints. Builds end-to-end products from concept to production — active hackathon participant with a track record of shipping under pressure.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Rust

AI/ML & LLM Engineering: Groq (Llama 3/3 70B), Gemini 1.5 Pro Vision, LLM integration & evaluation, RAG, pgvector semantic search, prompt engineering, NLP, TF-IDF, scikit-learn, embeddings

Frontend: Next.js, React, TypeScript, Tailwind CSS, Framer Motion, PWA/offline-first design

Backend & APIs: FastAPI, Node.js, Express, REST API design, BullMQ/Redis (async queuing), JWT/Bcrypt auth, asyncio

Databases & Infra: PostgreSQL, Supabase, SQLite, NeDB, Docker, Vercel, Row Level Security (RLS)

Testing & Practices: Git, pytest, Jest, CI-friendly test suites, system design, API-first architecture

PROJECTS

HealthLens — *Next.js 14, TypeScript, Groq, Gemini 1.5 Pro Vision, Supabase, Clerk, PWA* **2024**

- Built a multilingual AI healthcare PWA for rural/underserved users – shipped and demoed at Techflix hackathon as a fully deployed product.
- Engineered a deterministic Red Flag Emergency Engine that intercepts critical symptoms locally (e.g., chest pain, breathing difficulty) and bypasses the LLM entirely for zero-latency emergency redirection.
- Integrated Groq (Llama 3) for real-time symptom reasoning and Gemini 1.5 Pro Vision for image-based skin condition analysis; added voice input/output and 7-language support for accessibility.
- Implemented offline-first PWA with service workers and installable home screen support; Supabase Row Level Security (RLS) for HIPAA-ready data architecture.

SmartNews — *Python, FastAPI, scikit-learn, React, Docker, SQLite, Vercel* | *smartnews-tau.vercel.app* **2025**

- Built a production-grade news aggregation system scraping 5 sources (BBC, CNN, ToI, The Verge, HN) with automated 30-minute background scheduling via asyncio.
- Implemented full ML pipeline: TF-IDF + Logistic Regression topic classifier (7 categories), binary clickbait detector, and cosine similarity duplicate detection at 0.9 threshold.
- Designed personalization engine building per-user category preference vectors from interaction history; 7 REST endpoints via FastAPI. Containerized with Docker, deployed on Vercel.

MindFlow — *React, Node.js, Express, Groq (Llama 3 70B), Framer Motion, JWT, NeDB* | *smartstudy-nu.vercel.app* **2025**

- Built a full-stack AI study assistant that parses uploaded PDFs and text files and auto-generates interactive flashcards and multiple-choice quizzes using Groq (Llama 3 70B).
- Implemented flip-card animations with Framer Motion, session persistence, JWT/Bcrypt auth; deployed live on Vercel with a React + Vite frontend and Node/Express backend.

ReplySync — *Next.js, Node.js, WhatsApp Business API, Supabase, PostgreSQL, Razorpay* **2025**

- Built a WhatsApp automation CRM for Indian coaching institutes to manage leads, automate follow-ups, and streamline student communication at scale.
- Designed multi-tenant architecture with role-based access, automated drip campaigns, and template-based bulk messaging compliant with WhatsApp Business API policies.
- Built an analytics dashboard tracking response rates, conversion funnels, and campaign performance; integrated Razorpay for subscription billing.
- Engineered a queue-based message dispatch system to handle high-volume sends while respecting WhatsApp rate limits and avoiding account flags.

EDUCATION

Dayananda Sagar University

B.Tech, Computer Science and Engineering

Bengaluru, Karnataka, India

Aug 2025 – Aug 2029

ADDITIONAL

Hackathons: Techflix (HealthLens – shipped & deployed), BNMIT Virtuovation 2.0 (AR/VR), DSU internal hackathons.

Interests: AI infrastructure, LLM evaluation, developer tooling, system design.